DATA ACQUISITION PRESENTATION

SAM smart air measure

Group 7

DAQ + DA project

GROUP MEMBERS

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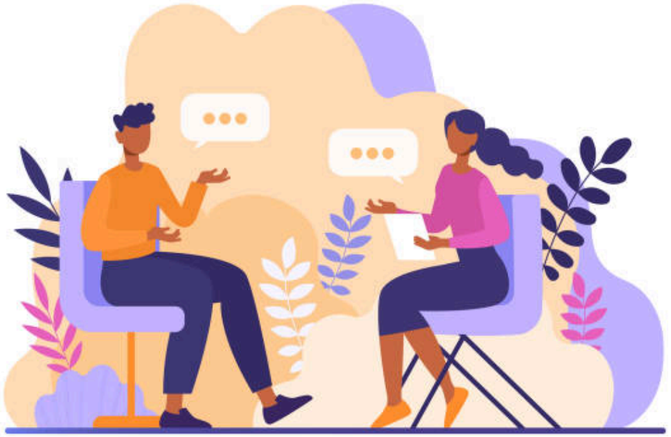


DAQ

Background and Movitation

Because PM2.5 pollution in Thailand has become severely hazardous to public health, the Smart Air Measure (SAM) project was created to help citizens stay constantly updated on their local air quality conditions. By combining real-time hardware sensor data with external weather APIs and machine learning models, SAM empowers users with both accurate historic insights and predictive forecasting.

Overall Architecture

IoT / Data Ingestion

- MicroPython on KidBright — sensor firmware
- mqtt — MQTT publish/subscribe protocol
- Node-RED — visual flow to parse MQTT → SQL INSERT → MySQL

Backend API

- FastAPI — async Python REST API framework
- SQLAlchemy + PyMySQL — database ORM
- httpx — async HTTP client for external API calls
- scikit-learn — ML model training and inference
- joblib — model serialization (`.pkl` files)

Overall Architecture

Frontend Web App

-Next.js + React + TypeScript
- App Router — `/features`, `/predict`, `/dashboard` pages
- CSS glass-morphism — blur backdrop, gradient headings

Data Analysis

- pandas — data wrangling and merging
- numpy — numerical operations
- matplotlib / seaborn — visualization
- Jupyter Notebook — interactive analysis environment

Data Collection and Data Sources

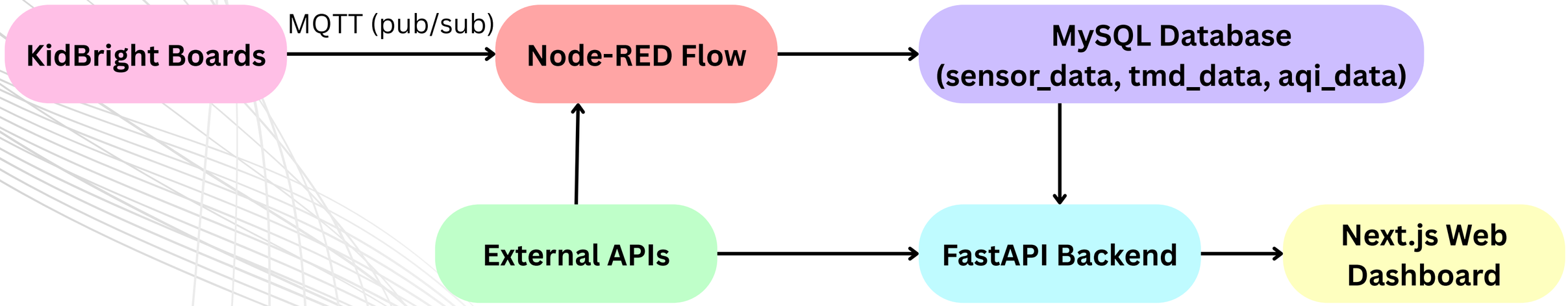
Primary Source

- Device: KidBright IoT Board.
- Method: Real-time data collection via external sensors.
- Measured Parameters:
  - Air Quality: PM2.5 and PM10 concentration.
  - Environmental: Ambient Temperature and Relative Humidity.
  - Chemical: Gas CO levels (Carbon Monoxide).
- Data collected firsthand from both Indoor and Outdoor environments to capture localized variations.

Secondary Source

- Source: TMD and AQI Web API.
- Method: Data extraction and integration via backend services.

Data Pipeline



IoT Data Pipeline

- **Protocol: MQTT (Hardware-to-Cloud communication).**
- **Hardware: KidBright boards integrated with MQ-2, MQ-9, and PMS7003 sensors.**
- **The Flow:**
  - **Publish: Hardware sends raw JSON packets via MQTT.**
  - **Process: Node-RED subscribes to topics and transforms raw voltages into clean metrics.**
  - **Store: Automated SQL insertion into the central MySQL database.**

Database schema used for data integration

sensor_data

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	**id**	int			No	*None*		AUTO_INCREMENT
2	**ts**	timestamp			No	CURRENT_TIMESTAMP		DEFAULT_GENERATED
3	**place**	varchar(20)	utf8mb3_general_ci		No	*None*		
4	**temp_dht**	float			No	*None*		
5	**temp_ky**	float			No	*None*		
6	**humidity**	int			No	*None*		
7	**pm10**	int			No	*None*		
8	**pm25**	int			No	*None*		
9	**pm100**	int			No	*None*		
10	**co_raw**	int			No	*None*		
11	**smoke_raw**	int			No	*None*		
12	**lat**	float			No	*None*		
13	**lon**	float			No	*None*		

Database schema used for data integration

aqi_data

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	**id**	int			No	*None*		AUTO_INCREMENT
2	**ts**	timestamp			No	CURRENT_TIMESTAMP		DEFAULT_GENERATED
3	**pm10**	int			No	*None*		
4	**pm25**	int			No	*None*		
5	**lat**	float			No	*None*		
6	**lon**	float			No	*None*		

tmd_data

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	**id**	int			No	*None*		AUTO_INCREMENT
2	**ts**	timestamp			No	CURRENT_TIMESTAMP		DEFAULT_GENERATED
3	**temp**	float			No	*None*		
4	**humidity**	float			No	*None*		
5	**rainfall**	float			No	*None*		
6	**lat**	float			No	*None*		
7	**lon**	float			No	*None*		

API sharing

1. **External Data Proxies**

Location & Contextualization

- **API: GET /api/location**
- **Function: Automatically detects location via User IP.**
- **Data: Resolves City, Coordinates (Lat/Lon), and Timezone.**
- **Benefit: Enables instant localization—no manual address entry needed.**

Environmental Intelligence

- **Weather API: GET /api/weather (via Open-Meteo)**
  - **Fetches UV Index, Rain probability, and 7-day forecasts.**
- **AQI API: GET /api/aqi (via AQICN)**
  - **Retrieves live Air Quality Index and specific pollutants (PM2.5, PM10, NO2, and O3.).**
- **Security: Acts as a Backend Proxy to hide sensitive API keys and bypass browser CORS restrictions.**

API sharing

Machine Learning Predictions

Live Data Integration

- **API: GET /api/predict/latest**
  - **Function: Bridges MySQL sensor data with external weather (TMD).**
  - **User Benefit: Instantly auto-fills the prediction form with real-time live data—eliminating manual entry.**

AQI Prediction Engine

- **API: POST /api/predict**
  - **The Input: Processes 9 key features (Temp, Humidity, CO, PM2.5, PM10, etc.).**
  - **The Logic: Executes two models simultaneously:**
    - **Random Forest (RF)**
    - **Multiple Linear Regression (MLR)**
  - **The Output: Real-time AQI results that drive the dynamic colored gauges on the dashboard.**

API sharing

Dashboard & Data Ingestion
System Health Monitoring

- **API: GET /api/dashboard/stats**
  - **Function: Verifies the connection between the FastAPI backend and the MySQL database.**
  - **Live Metrics: * Returns Connection Status (True/False).**
    - **Displays the Total Record Count of ingested sensor data.**
  - **User Benefit: Provides immediate confirmation that the system is live and data is being stored.**